

Inside sub_classic.csv

This document explains exactly what happens inside submissions/sub_classic.csv, which scored 89 on the Flipkart Gridlock leaderboard ($R^2 = 0.89$). It is a pure machine-learning pipeline: no lookup copy, no blend with Day-48 exact-match values. Every prediction is produced by a trained model from engineered features.

1. At a Glance

- Output file: submissions/sub_classic.csv (41,778 rows)
- Leaderboard score: 89 ($R^2 = 0.89$)
- Model inside: ExtraTreesRegressor (scikit-learn ensemble of randomized trees)
- Training data: Day 48 only (69,427 rows); test window Day 49 hours 2:15-13:45
- Feature count: 35 numeric engineered features + 4 one-hot RoadType columns = 39 inputs
- Scripts: classic_ml.py (tournament), submit_classic.py (writes CSV), src/classic_features.py

2. End-to-End Pipeline Flow

```
dataset/train.csv + dataset/test.csv
|
v
src/pipeline.py -> clean, decode geohash, cyclical time, categoricals
|
v
Day-48 aggregates (hist_g, hist_t, hist_g5, hist_g5t) + L00 on train rows
|
v
src/spatial.py -> hist_nb (12-neighbor spatial prior at same minute)
|
v
src/classic_features.py -> geohash profiles + interactions + test-band flags
|
v
to_design_matrix() -> 39-column numeric matrix X
|
v
ExtraTreesRegressor.fit(X_train, demand) on all Day 48 rows
|
v
predict(X_test) -> clip to [0, 1] -> sub_classic.csv
```

3. Data Preprocessing (shared base)

Both train and test rows pass through src/pipeline.py before feature engineering.

- Parse timestamp to minutes since midnight
- Cyclical encoding: $\min_sin, \min_cos = \sin/\cos(2\pi * \text{minutes}/1440)$
- Decode 6-char geohash to lat/lon; extract 5-char prefix g5
- Impute Temperature with train median; NumberofLanes missing -> 0
- Encode RoadType, LargeVehicles, Landmarks, Weather as integers

- All historical stats computed from Day 48 only (no Day 49 labels in training)

3.1 Leakage-Safe History Features

- hist_g: leave-one-out mean demand per geohash (Day 48 train rows)
- hist_t: LOO mean demand per timestamp (global time-of-day curve)
- hist_g5: LOO mean per geohash prefix
- hist_g5t: LOO mean per (g5 prefix, timestamp)
- For test rows: plain Day 48 means (no LOO needed)

hist_gt (exact Day-48 slot copy) is deliberately NOT used in this pipeline.

3.2 Spatial Prior (hist_nb)

- KDTree over Day 48 geohash coordinates
- k=12 nearest neighbors (excluding self)
- Mean neighbor demand at same minute; +/- 45 min window if sparse
- Fallback: hist_g, then global mean

4. Classic Feature Engineering (35 features)

src/classic_features.py builds interpretable groups. Each group targets a different slice of demand variance.

4.1 Location Group (14 features)

- lat, lon - decoded coordinates
- hist_g, hist_g5, hist_nb - Day-48 spatial priors
- d48_std, d48_max, d48_min, d48_range - per-geohash demand spread on Day 48
- d48_morning - mean demand 0:00-2:00 at this geohash
- d48_testhrs - mean demand 2:15-13:45 at this geohash
- d48_amp - ratio testhrs/morning (daytime uplift factor)
- d48_peak_min - minute of peak demand on Day 48
- log_hist_g - $\log_{10}(\text{hist_g})$ for skewed demand scale

Exploits ~99% geohash overlap: we have seen almost every test location on Day 48.

4.2 Time Group (6 features)

- minutes, min_sin, min_cos - time of day
- hist_t - global time-of-day prior from Day 48
- dist_peak - $|\text{minutes} - \text{d48_peak_min}|$ (distance from location peak)
- in_test_band - 1 if timestamp in 2:15-13:45, else 0

4.3 Road Group (3 numeric + 4 one-hot)

- NumberofLanes, LargeVehicles, Landmarks (numeric)
- RoadType one-hot: Residential(0), Street(1), Highway(2), Unknown(3)

Road type is the strongest categorical split in the data.

4.4 Weather Group (1 feature)

- Temperature (imputed median)

Weather adds almost no incremental variance in our analysis.

4.5 Interaction Group (7 features)

- ix_g_sin = $\text{hist_g} * \text{min_sin}$
- ix_g_cos = $\text{hist_g} * \text{min_cos}$
- ix_g_t = $\text{hist_g} * \text{hist_t}$ (location-specific time effect)
- ix_t_sin = $\text{hist_t} * \text{min_sin}$
- ix_g_rt = $\text{hist_g} * \text{RoadType}$
- ix_g_band = $\text{hist_g} * \text{in_test_band}$
- ix_t_band = $\text{hist_t} * \text{in_test_band}$

These make non-linear relationships explicit for linear models and give trees additional split cues for location-time-road combinations.

5. How Features Explain Variance

$R^2 = 1 - (\text{unexplained variance} / \text{total variance})$. We measured incremental R^2 on the testhours holdout (train Day 48 outside 2:15-13:45, predict that band) using Ridge regression to show what each group contributes.

- +location cumulative R2 = 0.5573 (14 features)
- +time cumulative R2 = 0.6099 (20 features)
- +road cumulative R2 = 0.8224 (27 features)
- +weather cumulative R2 = 0.8224 (28 features)
- +interactions cumulative R2 = 0.8593 (35 features)

Interpretation: demand variance is mostly between locations (hist_g, profiles), then road type, then time-of-day interactions. Weather is negligible.

6. The Model: ExtraTreesRegressor

sub_classic.csv uses Extremely Randomized Trees - a classic bagging ensemble. Each of 400 trees splits on random feature subsets; predictions are averaged.

6.1 Hyperparameters

- n_estimators = 400
- max_depth = 16
- min_samples_leaf = 10
- max_features = sqrt(n_features)
- random_state = 0

No log transform on target. Predictions clipped to [0, 1] after predict().

6.2 Why ExtraTrees (not Ridge or LightGBM)?

classic_ml.py ran a tournament on the same 39-feature matrix:

- ridge testhours R2 = 0.8593
- extra_trees testhours R2 = 0.8427
- elasticnet testhours R2 = 0.8403
- gbm_raw_features testhours R2 = 0.8400
- random_forest testhours R2 = 0.8238
- lgbm_classic testhours R2 = 0.7889
- hist_gbm testhours R2 = 0.7826

Winner selection: $0.6 * \text{testhours_R2} + 0.4 * \text{random_R2} \rightarrow \text{extra_trees}$. ExtraTrees balanced strong testhours skill (0.8427) with high random-CV (0.9467). Ridge had the highest testhours R2 (0.8593) but ExtraTrees won the combined score.

7. Training and Inference (submit_classic.py)

7.1 Training

- Load train.csv and test.csv
- Preprocess all rows via prepare()
- Build Day-48 aggregates, spatial index, geohash profiles
- Engineer 35 features on Day 48 training rows (with LOO hist features)
- Build design matrix X (39 columns) and target y = demand
- Fit ExtraTreesRegressor on all 69,427 Day-48 rows

7.2 Inference (test rows)

- Same preprocessing and feature engineering on test.csv
- hist_* features use plain Day-48 means (not LOO)
- Geohash profiles from full Day 48
- model.predict(X_test) for 41,778 rows
- Clip predictions to [0, 1]
- Write Index + demand to sub_classic.csv (Index order matches test.csv)

7.3 What Is NOT Inside sub_classic.csv

- No lookup ladder (no copying Day-48 demand at same geohash+timestamp)
- No hist_gt feature
- No blend with lookup (W_LOOKUP = N/A)
- No Day 49 morning labels in training
- No cross-day stack model

8. Comparison to Other Submissions

- sub_gbm.csv (LightGBM, 16 raw features): LB score 90, R2 = 0.90
- sub_classic.csv (ExtraTrees, 39 engineered features): LB score 89, R2 = 0.89
- sub_classic_ridge.csv (Ridge, same features): LB score 88

The classic pipeline scores 89 - one point below the notebook LightGBM. More features and variance-aware engineering helped vs raw Ridge (88), but LightGBM on the simpler 16-feature set still edges ahead on the leaderboard. Trees on the engineered matrix capture road-type and interaction effects well, but boosting on raw hist features generalizes slightly better to Day 49 midday.

9. Local Validation Scores

Scores from classic_ml_report.json (leak-safe holdouts):

TESTHOURS

- ridge: R2 = 0.8593
- extra_trees: R2 = 0.8427 <-- winner
- elasticnet: R2 = 0.8403
- gbm_raw_features: R2 = 0.8400

- random_forest: R2 = 0.8238
- lgbm_classic: R2 = 0.7889
- hist_gbm: R2 = 0.7826

RANDOM

- lgbm_classic: R2 = 0.9537
- random_forest: R2 = 0.9495
- hist_gbm: R2 = 0.9491
- extra_trees: R2 = 0.9467
- gbm_raw_features: R2 = 0.9294
- ridge: R2 = 0.9112
- elasticnet: R2 = 0.9054

D49_MORNING

- extra_trees: R2 = 0.8060
- random_forest: R2 = 0.7904
- gbm_raw_features: R2 = 0.7709
- lgbm_classic: R2 = 0.7580
- ridge: R2 = 0.7384
- hist_gbm: R2 = 0.7369
- elasticnet: R2 = 0.7150

Testhours holdout (2:15-13:45 on Day 48) is the bracket most aligned with the real test window. Morning holdout (0:00-2:00) is a weaker LB proxy.

10. Key Files

- src/pipeline.py - base cleaning and Day-48 aggregates
- src/spatial.py - neighbor prior hist_nb
- src/ml_features.py - profile stats and base interactions
- src/classic_features.py - full 35-feature engineering + design matrix
- classic_ml.py - model tournament and variance report
- submit_classic.py - trains winner and writes sub_classic.csv
- submissions/classic_ml_report.json - full tournament results

11. How to Reproduce

```
python classic_ml.py
python submit_classic.py
```

Output: submissions/sub_classic.csv ready for manual upload to the competition platform.